Public Trust Confidential security Clearance — Current

Reno Rajan Christy

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Senior Data & AI Engineering Professional • 14+ Years in Banking & Finance Technology

• **Microsoft Fabric** • **Azure Synapse** • **Azure Databricks** • **Snowflake** • **Generative AI** • **Cloud Data Architecture**

Senior Data & AI Engineering professional with 14+ years delivering enterprise cloud data platforms across banking, payments, credit risk, and fraud analytics. Deep expertise in Data engineering, Microsoft Fabric, Azure Synapse, Azure Databricks, and Snowflake building Medallion architectures, real-time pipelines, and GenAI/LLM integrations (OCI GenAI, Azure OpenAI, RAG). Proven at U.S. Bank and J.P. Morgan Chase across AML/fraud decisioning, CI/CD data platforms, and regulatory compliance. Strong Python, PySpark, SAS, and SQL skills with hands-on Airflow and enterprise data governance. Experienced in Credit Risk, Core banking solutions, Payments, Wealth Data Solutions and Intelligent Automation for Enterprise Financial Systems.

CORE COMPETENCIES

Advanced ETL, Python, Data Modelling, Data Pipelines, Data Warehousing, real time data streaming, Artificial Intelligence solutioning (Azure), solutions architecture and redesigning, business process automation, technology management and governance, product ideation, implementation, dependency solutioning and value evolution, Exploratory Data Analysis, Predictive Modeling, Time series Forecasting, Machine Learning

EDUCATION

Master of Science in Management Information Systems – (Data Science focus) May 2019
Oklahoma State University, Stillwater, Oklahoma, USA GPA 4.0 / 4.0

Bachelor of Technology in Electronics and Communication Engineering May 2012
(Distinction graduate), Mahatma Gandhi University, Kerala, India GPA 3.81/4.0

PROFESSIONAL CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Artificial Intelligence Professional
- Oracle Cloud OCI Certified Artificial Intelligence Associate
- Microsoft Certified Azure Data Engineer Associate DP203
- Microsoft Certified Artificial Intelligence AI 900
- Microsoft Certified Azure fundamentals AZ900
- SAS Certified Advanced SAS Programmer
- SAS Certified Base SAS Programmer
- Certified Associate of Banking & Finance (CAIIB – IIBF)
- Junior Associate of Banking & Finance (JAIIB – IIBF)
- Data camp Cert: Python Data Analyst-career track
- Data camp Cert: R programmer skill track
- Data camp Cert: Multiple Python skill tracks
- edX: Analyzing and Visualizing Data - Power BI
- Architecting in AWS & AWS technical
- Udemy Cert: Tableau A-Z
- Data Analysis with Adobe Analysis workspace
- AWS Certified Cloud Practitioner
- Microsoft Cert Data fundamentals DP 900
- ISTQB Certified software tester
- Tableau Certified Desktop Specialist

SKILLS & EXPERTISE

Data Engineering: Azure Fabric, ETL Development, Data Pipeline Design, Data Warehousing, Data Governance, IoT
Programming: SQL, SAS, SSIS, ETL, pyodbc, sklearn, requests, Duckdb, Python (Pandas, PySpark, PyArrow), R, Scala, bash
Databases & Tools:RDBMS, Data Lake, ODS, Hive, Teradata, Oracle, DB2, MS SQL Server
Analytics & Reporting: Tableau, QlikView, A/B testing, Adobe Analytics, Power BI, Steamlit, Splunk, Grafana
Cloud & Big Data Platforms: Azure (ADF, Synapse, Databricks, Fabric), AWS(S3, Glue, Athena, Redshift, EventBridge, SNS, SQS, AWS CLI), GCP (Big Query), Snowflake, Delta Lake, Data Lake Storage), Hadoop, Snowflake, Snowpipe, Snowpark,
Orchestration & Workflow Management: Apache Airflow, Prefect, Azure Data Factory, Azure DevOps, Github actions
Data Modeling & Architecture: Dimensional Modeling, Star/Snowflake Schemas, ERD Modeling, 3NF Normalization, SCD
Data Quality & Monitoring: Validation models, Schema Enforcement, Deduplication, Anomaly Detection, Alerting Dashboards
Version Control & CI/CD: Git, GitHub, Azure Repos, CI/CD, dbt Cloud, YAML-based deployment automation
Decisioning Platforms: Fraud Detection, Credit Line, Marketing Campaigns, Risk Management
Collaboration: Business Analysis, Cross-functional Team Leadership, IT-Risk Coordination
Technology Integration:Mainframe, Middleware, API ingestion python, multiplatform architecture integration
Industries: Banking, Finance, Marketing Analytics, Software and Strategy Consulting
ML & AI: Feature Engineering, Decision trees, Random forest, logistic regression, NLP

PROFESSIONAL EXPERIENCE

Sr Data Engineer, U.S. Bank (FT+CT), Minneapolis, USA **Oct 2022 to Present**

Handled key responsibility of maintaining and enhancing critical enterprise business solutions for Real Time Credit Card Authorizations application for the entire banks credit card portfolio. Designed and architected critical backend data architecture for Analytical solution supporting corporate credit cards payment platform affecting multiple key corporate client organizations and MNC's including federal agencies.

- Built and optimized Fabric Notebooks and Spark applications using Python, PySpark, and SQL to process, validate, and transform large datasets, enabling advanced analytics and seamless data integration within Lakehouse environments.
- Designed and implemented end-to-end Microsoft Fabric architecture leveraging OneLake, Lakehouse, Fabric Warehouses, and Power BI semantic models to support real-time and analytical banking workloads.
- Implemented Medallion Architecture (Bronze/Silver/Gold) in Fabric Lakehouse to standardize ingestion, cleansing, business transformations, and reporting layers across payments and risk analytics domains.
- Physical-to-cloud IoT intelligence solution integrating industrial IoT edge-to-cloud solution with firmware sensor nodes, Azure IoT Hub ingestion, ADLS Lake house achieving 95% cost reduction.
- Used OCI AI for prompt engineering, fine-tuning small models, and integrating GenAI into Python data pipelines. Used OCI embeddings and vector search for semantic similarity and information retrieval.
- SAS programming, leveraging advanced data manipulation and ETL techniques to drive data-driven decision-making for reporting and analytical operational data store needs including Fraud rules and strategies, payments analytics and credit decisions.
- Developed and optimized SAS macros, Proc SQL queries, and ETL jobs to automate fraud rules, payment and financial controls.
- Spearheaded the enhancement of a high-availability enterprise solution for real-time card authentications, integrating technologies like mainframe, IBM MQ, TIBCO, DB2, Python, js, and Kafka deployed in a robust internal cloud environment with CICD in dockerized environment.
- Created and modified several database objects such as Tables, Views, Indexes, Constraints, Stored procedures, Packages, Functions and Triggers using T SQL and other platforms
- Developed queries, streams, and tasks in the Snowflake cloud data platform, enabling warehousing, analytics data pipelines.
- Implemented Snowflake-native components such as Snowpipe for real-time data ingestion, Streams and Tasks for change data capture (CDC), enabling scalable and efficient analytics workflows across finance and fraud

Key areas: Azure Data Engineering, Azure Synapse, Fabric, Data Factory, Power BI, Azure Cloud Data Solutions Architecting, data modelling, data engineering, reporting, python, Splunk alerts, analytics, Public Trust Federal security clearance, Snowflake, Databricks, IoT Hub, Industrial firmware to cloud streaming

Sr Data Engineer, Quantitative Analytics, J P Morgan Chase Bank, Seattle-USA **Jan 2022 to Oct 2022**

Principally spearheaded the data feeds design for product migration of Customer Journey Analytics platform from Bryter CX to Adobe Customer Journey Analytics. Designed and implemented the ETL feeds for multiple delivery channel platforms such as IVR, Card systems, Surveys, ATMs, Tellers etc. based off of multiple platform solutions based off of Teradata, MySQL, Oracle, Big Data Hortonworks Hive etc. specific to the independent feed source origin platforms.

- Designed and developed ETL data feeds for key banking operations, including Teller, ATM, Card-IVR, Surveys, Alerts, Card Fees, and Online Transactions, leveraging tools like Teradata, Oracle, Hive, and Python. Bryter Cx to Adobe CJA product migration.
- Led the architectural redesign and implementation of data pipelines and ETL processes, facilitating seamless migration to Adobe Customer Journey Analytics.
- Automated data extraction processes using Python for seamless integration with Adobe API interfaces.
- Guided and trained members on multi-platform ETL solutions, enabling Adobe Customer Journey Analytics platform. IVR forecast volume prediction estimation using Sarima, evaluation for bias and trend, pattern trending, using Python, PySpark

Data Engineer II, Credit One Bank, NV, USA **Aug 2019 to Jan 2022**

Developed data engineering ETL solutions for various business processes such as marketing campaigns, Card acquisition strategies development, engineering features for decisioning platforms etc. Spearheaded the Data Quality monitoring and improvement of Data lake systems responsible for critical decisioning systems such as Credit Line increase, Red Flag application monitoring, Marketing campaigns, Credit Card Acquisitions, Credit Risk measurements.

- Maintained and optimized a complex RDBMS Data Lake-ODS ecosystem, managing over 700 stored procedures and supporting critical decisioning platforms processing $3 billion annually. Led weekly data triages for 9 months to enhance reliability.
- Conducted rigorous data quality improvements, implementing 180+ major fixes in a production ecosystem with 3,600+ tables and 900+ stored procedures, improving accuracy in fraud, credit line systems.
- Engineered Python-based data solutions, including 7 major projects for business process automation and data analytics, projects in Marketing Campaigns, card acquisitions, audits. Py pandas, numpy, pyodbc, smptlib, os, scipy, sklearn etc.
- Developed analytical solutions for fraud and acquisitions teams using SAS, SQL, Python and Tableau
- Validated data implementations for ODS databases, ensuring accuracy and compliance for risk management operations.
- Supported Marketing segmentation analysis, loss forecasting, charge off estimations, behavior scoring inputs, credit policy evaluation analysis and adjustment support activities, Credit line increase data pipeline maintenance, applicant fraud flagging.
- Developed data ware house-based business reports based on the dimensional data warehouse platform using SSIS and T SQL
- Engineered Python-based automation solutions to enhance data-driven marketing, credit risk assessment, and fraud detection strategies.
- Designed and developed 45 ETL solutions end to end for marketing campaign analytics initiatives
- Collected, curated, and documented data definitions as part of a data governance initiative to ensure consistency.
- Maintained the Critical Data Elements and features engineering for various Risk, portfolio decisioning systems

Key areas: SQL, SSIS, SAS, Data warehousing, Data Governance, Data Quality, statistical modelling, Risk and credit Analytics

Summer IT intern, JBS USA (Swift Beef Co), Colorado, USA **June 2018 – Aug 2018**

- QlikView Application Development for SAP license cost reductions and mapping of IT Architecture systems

Tools : Qlikview Application Development and Reporting, Datawarehouse Reporting with SQL, Visio IT Architecture mapping

Graduate Teaching Assistant, Oklahoma State University, USA **Jan 2018 - June 2019**

Key areas: Python, R, SQL, Statistical Modeling, Time Series, Regressions, Decision Trees, Knime, Data Science, Machine learning, Business Intelligence Tools, Knime Decision Science modelling, Economic analysis, Research

- Programming in Data-science with R and Python – GA@ Masters Level
- Business Intelligence and Predictive Analytics (with Data Mining) – GA@ Masters Level
- PTSD Health project – Oracle Cerner EMR systems, Integrated Cerner EHR and patient data to support healthcare analytics use cases, including PTSD/behavioral health risk indicators.

Assistant Manager IT & Specialist IT Officer, Canara (Syndicate) Bank, India **July 2014 – Oct 2016**

- Primary liaison and reporting officer for IT risk assurance, regulatory reporting, and compliance for the Syndicate National bank's IT systems, working with Nation's Central bank executives of Reserve Bank of India for Risk Based Supervision on assessment of Risk and Capital (SPARC) Central Bank regulatory program - Supervisory Program on Assessment of Risk and Capital

- Led technology and data reporting efforts, focusing on non-performing assets and Risk, Audit reporting
- Developed SQL-based reports for regulatory compliance, MIS, branch and executive reporting and business analysis
- Lead the technical development of banking business data analysis reports for performing competitive analysis for future Bank growth such as deposit trends, credit growth and portfolio trends.
- Lead the process improvement efforts on review, revision and formulation of various policies and strategies of the Bank resulting in approx. 2 Million on monetary value benefit
- Communication, Coordination and its executive decisioning responsibility for international business branch divisions on matters related to country specific regulatory needs, compliance, reporting, Commercial and multimillion Corporate syndicated loans
- Developed Analytical reporting solutions for agricultural loans loss mitigation, commercial provisioning requirements on Non-Performing Assets, Gold loans, Debt restructuring opportunities and Risk mitigations
- Worked extensively with Leadership including CIO, CISO, Executive technical leaders and corporate divisions on matters of business reporting, compliance, risk, RCSA, technology risk assessments and vendor application management.
- Lead enhancement implementation, upgrades improvement for 7 ancillary apps integrated to the Banks Core Banking Solution, Designed and supported solutions integrating ancillary platforms for structured, interoperable data exchange ensuring seamless financial messaging and transaction processing.
- Executed IT Procurement related activities like High Value Financial sanctions, Feasibility, Impact assessment, cross functional dependency, multi-vendor project delivery aspects, BCP, RFP Scrutiny, Legal clauses and AMC/ATS bifurcations. (IT procurement of USD 70 million per annum)
- Executed the decision making for Audit issue resolution of Management/ Information Security/ Concurrent Audits and Operational Risk Management Points, ABC Audit Analysis, Risk Control and Self-Assessment Points with external vendors.
- Managed various regulatory reporting aspects across capital markets, fixed-income instruments, equities, portfolio adjustments, NPAs, contract reporting, liabilities, assets, and liquidity adjustments, ensuring compliance with financial regulations.

Software Engineering Analyst, Accenture, India (Clients KPN, Telus Canada) Jan 2013 to May 2014

- Developed data warehouse reporting and QA solutions using SSIS, SQL, SSAS MDX, DMX, QTP, QC ALM
- Programmed ETL Queries to support development and architect team for data vault and staging implementations
- Coded end to end data flow from multiple Data sources to staging, staging to data warehouse (data vault model), Warehouse to data marts (dimension) and Data Marts to OBIEE for QA
- Performed Data warehouse reporting in SSAS using MDX and using T SQL for OLTP reports
- Analyzed business requirements, technical specifications, technical design docs for defects, RCA and performed test planning, execution, defect tracking, incident reporting etc. in QC/ALP App for IVR-CTI projects in telecom domain

Key areas: BI Development, SQL, Data warehouse Reporting, SSIS, SSAS, SQL, ETL development, QA Validation, MDX, DMX, reporting